

E-Commerce Sales and Customer Intelligence Dashboard

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1 E-Commerce Business Intelligence Dashboard as a real portfolio project.

We'll do it in stages:

1. Define the business problem and requirements
2. Create the e-commerce dataset/database
3. Load data into MySQL
4. Perform advanced SQL analysis
5. Clean/transform data with Power Query
6. Build a proper Star Schema
7. Create DAX measures
8. Build the Power BI dashboard
9. Extract business insights and recommendations

1.1 Business scenario

Me as a Data Analyst for an e-commerce company and the management team wants answers to questions such as:

- How much revenue are we generating?
- Which products and categories perform best?
- Which customers generate the most revenue?
- Which regions are performing well?
- Are sales increasing or decreasing?
- What is our profit margin?
- What is the average order value?
- How many customers are returning?
- Which products have poor profitability?
- How is performance changing month-over-month?

1.2 Our technology stack

We'll deliberately use the technologies you want to strengthen:

So by the end, I will be able to say I built an end-to-end BI solution starting from relational data in MySQL, performed advanced SQL analysis, transformed the data using Power Query, created a star-schema model, developed DAX measures and built an interactive Power BI dashboard.

1.3 Business requirements

this is an important part that most student Power BI projects skip.

We'll create a small Business Requirements Document.

Stakeholders

E-Commerce Management

1.3.1 Business objective:

Develop an interactive business intelligence dashboard that enables management to monitor sales, profitability, customer behavior and product performance.

1.4 Key business questions:

- Sales
 1. What is total revenue?
 2. What is total profit?
 3. What is the monthly sales trend?
 4. What is the month-over-month growth?
- Customers
 1. Who are our highest-value customers?
 2. What percentage of customers are returning?
 3. Which customer segments generate the most revenue?
- Products
 1. Which products generate the most revenue?
 2. Which products generate the most profit?
 3. Which categories have poor margins?
- Geography
 1. Which regions generate the most revenue?
 2. Which stores perform best?
- Management
 1. What are our key KPIs?
 2. Where are the biggest opportunities for improvement?

1.5 Define our KPIs:

KPI	Meaning
Total Revenue	Total sales generated
Total Profit	Revenue - cost
Profit Margin	Profit as % of revenue
Total Orders	Number of orders
Total Customers	Unique customers
Units Sold	Total quantity sold
Average Order Value	Revenue / orders
MoM Growth	Month-over-month revenue growth

We're going to build the underlying data first.

1.6 Dataset

For this project, I recommend that I create our own realistic dataset rather than downloading a random dataset.

That gives us control over:

- SQL relationships
- Data quality issues
- Missing values
- Duplicate records
- Dates
- Customers
- Products
- Stores
- Revenue
- Cost
- Discounts

And it lets us deliberately create a dataset that demonstrates the skills employers are looking for.

1.6.1 Target size:

Category	Count
Customers	1,000
Products	100
Stores	15
Orders	10,000+
Sales rows	10,000+

Enough data to make the dashboard realistic without making it unnecessarily difficult to work with.

2 First task:

I Should Required:

- MySQL
- MySQL Workbench
- Power BI Desktop

Recommended

- R/Python
- Git
- GitHub

I already have MySQL, Power BI, Python and R, so we're going to make those skills demonstrable through the project rather than just listing them.

3 Results:

3.1 Sales

Q: What is total revenue?

Total.Revenue
516092560

Q: What is total profit?

Total.Profit
64511985

Q: What is the monthly sales trend?

Month	Monthly.Revenue
2024-01	15459100
2024-02	14647485
2024-03	14872026
2024-04	17294594
2024-05	19968241
2024-06	17255695
2024-07	21490022
2024-08	19067253
2024-09	19863888
2024-10	21825825
2024-11	20788313
2024-12	21411570
2025-01	20887223
2025-02	22358141
2025-03	22726008
2025-04	22475136
2025-05	25148786
2025-06	24848495
2025-07	25917628
2025-08	25803207
2025-09	24090538
2025-10	26579703
2025-11	23804103
2025-12	27509580

Q:What is the month-over-month growth

Month	Revenue	Previous.MonthRevenue	MoM.Growth.Percent
2024-01	15459100	NULL	NULL
2024-02	14647485	15459100.24	-5.25
2024-03	14872026	14647484.52	1.53
2024-04	17294594	14872025.79	16.29
2024-05	19968241	17294593.53	15.46
2024-06	17255695	19968241.1	-13.58
2024-07	21490022	17255695.27	24.54
2024-08	19067253	21490021.55	-11.27
2024-09	19863888	19067253.47	4.18
2024-10	21825825	19863887.74	9.88

Month	Revenue	Previous.MonthRevenue	MoM.Growth.Percent
2024-11	20788313	21825825.08	-4.75
2024-12	21411570	20788313	3
2025-01	20887223	21411570.12	-2.45
2025-02	22358141	20887222.66	7.04
2025-03	22726008	22358141.03	1.65
2025-04	22475136	22726008.03	-1.1
2025-05	25148786	22475135.69	11.9
2025-06	24848495	25148785.9	-1.19
2025-07	25917628	24848494.84	4.3
2025-08	25803207	25917628.34	-0.44
2025-09	24090538	25803207.49	-6.64
2025-10	26579703	24090538.45	10.33
2025-11	23804103	26579703.01	-10.44
2025-12	27509580	23804103.14	15.57

3.2 Customers

Q: Who are our highest-value customers?

CustomerID	CustomerName	CustomerSegment	Total.Orders	Total.Revenue
CUST0635	Arjun Gupta	Consumer	14	1656986
CUST0964	Kavya Mishra	Consumer	12	1577143
CUST0642	Meera Rao	Consumer	12	1413467
CUST0979	Sneha Yadav	Consumer	14	1377343
CUST0495	Sameer Kumar	Consumer	10	1333987
CUST0894	Ishita Joshi	Consumer	12	1319199
CUST0683	Sameer Kumar	Consumer	14	1301036
CUST0146	Ishita Saxena	Consumer	14	1278361
CUST0860	Zoya Mishra	Corporate	12	1263996
CUST0108	Simran Joshi	Consumer	13	1202905
CUST0508	Mohammad Chauhan	Consumer	12	1173008
CUST0546	Simran Rao	Corporate	13	1162308
CUST0702	Sara Verma	Corporate	11	1149966
CUST0241	Sneha Bansal	Consumer	10	1148067
CUST0712	Neha Qureshi	Consumer	11	1148048
CUST0668	Aarav Agarwal	Consumer	11	1143479
CUST0896	Riya Qureshi	Consumer	13	1109516
CUST0671	Sara Srivastava	Consumer	10	1108307
CUST0920	Karan Rao	Consumer	9	1107595
CUST0604	Vikram Bansal	Small Business	9	1106282

Q: What percentage of customers are returning?

Customer.Type	Number.of.Customers	Percentage
Returning	993	99.4
One-Time	6	0.6

Q: Which customer segments generate the most revenue?

CustomerSegment	Customers	Revenue
Consumer	711	367703718
Corporate	171	86318092
Small Business	117	62070750

3.3 Products

Q: Which products generate the most revenue?

ProductID	ProductName	Category	SubCategory	Revenue	Units.Sold
PROD0061	Trail Equipment 61	Sports	Equipment	10980587	282
PROD0023	PageTurner Academic 23	Books	Academic	9884489	252
PROD0074	FitMax Equipment 74	Sports	Equipment	9673754	219
PROD0060	UrbanNest Kitchen 60	Home & Kitchen	Kitchen	9333555	253
PROD0076	BrightKids Children 76	Books	Children	9313601	266
PROD0001	UrbanWear Footwear 1	Fashion	Footwear	9172283	265
PROD0063	HomeEase Appliances 63	Home & Kitchen	Appliances	9101376	262
PROD0088	Zenith Laptops 88	Electronics	Laptops	9089534	272
PROD0002	HomeEase Kitchen 2	Home & Kitchen	Kitchen	9064444	221
PROD0049	Apex Smartphones 49	Electronics	Smartphones	8964110	248
PROD0030	LuxeCare Personal Care 30	Beauty	Personal Care	8963078	289
PROD0011	StyleHub Footwear 11	Fashion	Footwear	8910470	250
PROD0020	CookPro Furniture 20	Home & Kitchen	Furniture	8788921	271
PROD0078	FitMax Fitness 78	Sports	Fitness	8728222	275
PROD0048	Orbit Accessories 48	Electronics	Accessories	8648337	237
PROD0090	HomeEase Kitchen 90	Home & Kitchen	Kitchen	8559280	245
PROD0071	Bloom Haircare 71	Beauty	Haircare	8539845	246
PROD0007	Zenith Audio 7	Electronics	Audio	8445599	209
PROD0044	PageTurner Fiction 44	Books	Fiction	8306109	249
PROD0081	Trail Sportswear 81	Sports	Sportswear	8279102	270

Q: Which products generate the most profit?

ProductID	ProductName	Category	Revenue	Profit
PROD0074	FitMax Equipment 74	Sports	9673754	2042802
PROD0023	PageTurner Academic 23	Books	9884489	1963975
PROD0063	HomeEase Appliances 63	Home & Kitchen	9101376	1863780
PROD0076	BrightKids Children 76	Books	9313601	1716734
PROD0002	HomeEase Kitchen 2	Home & Kitchen	9064444	1678352
PROD0071	Bloom Haircare 71	Beauty	8539845	1628284
PROD0061	Trail Equipment 61	Sports	10980587	1578056
PROD0088	Zenith Laptops 88	Electronics	9089534	1537157
PROD0049	Apex Smartphones 49	Electronics	8964110	1528502
PROD0030	LuxeCare Personal Care 30	Beauty	8963078	1510915
PROD0094	Scholar Children 94	Books	7939877	1497912

ProductID	ProductName	Category	Revenue	Profit
PROD0060	UrbanNest Kitchen 60	Home & Kitchen	9333555	1477296
PROD0018	Trail Equipment 18	Sports	7231020	1394879
PROD0081	Trail Sportswear 81	Sports	8279102	1391456
PROD0058	Sprint Sportswear 58	Sports	7159900	1378380
PROD0042	Comfort Appliances 42	Home & Kitchen	6988762	1355619
PROD0007	Zenith Audio 7	Electronics	8445599	1337035
PROD0083	Trail Sportswear 83	Sports	7845541	1311847
PROD0078	FitMax Fitness 78	Sports	8728222	1282792
PROD0016	BrightKids Fiction 16	Books	6841764	1280363

Q: Which categories have poor margins?

Category	Revenue	Profit	Profit.Margin.Percent
Fashion	68515812	6183519	9.02
Electronics	82323666	9351043	11.36
Sports	107213430	13762322	12.84
Books	106267294	13752135	12.94
Beauty	51540409	7083384	13.74
Home & Kitchen	100231949	14379582	14.35

3.4 Geography

Q: Which regions generate the most revenue?

```
t11=read.csv("C:/Users/M.QasimSalman/Desktop/MSc/MYSQL/ecommerce/Answers/Region/Q1.csv")
kable(t11)
```

Region	Total.Orders	Revenue	Profit
North	2278	174852781	22330408
South	1798	136475003	16834515
West	1345	103316116	12832127
East	1329	101448661	12514935

Q: Which stores perform best?

```
t12=read.csv("C:/Users/M.QasimSalman/Desktop/MSc/MYSQL/ecommerce/Answers/Region/Q2.csv")
kable(t12)
```

StoreID	StoreName	City	State	Region	Total.Orders	Revenue	Profit
STORE002	Store 02	Noida	Uttar Pradesh	North	470	38518829	5046511
STORE004	Store 04	Jaipur	Rajasthan	North	484	36358610	4556396
STORE012	Store 12	Kochi	Kerala	South	474	35719186	4397173
STORE008	Store 08	Ahmedabad	Gujarat	West	475	35656083	4539546
STORE006	Store 06	Mumbai	Maharashtra	West	429	35587430	4402731
STORE014	Store 14	Patna	Bihar	East	432	34842276	4340370
STORE011	Store 11	Hyderabad	Telangana	South	442	34557494	4218327
STORE013	Store 13	Kolkata	West Bengal	East	466	33963016	4019699

StoreID	StoreName	City	State	Region	Total.Orders	Revenue	Profit
STORE001	Store 01	New Delhi	Delhi	North	452	33742761	4360099
STORE003	Store 03	Lucknow	Uttar Pradesh	North	443	33495936	4075986
STORE009	Store 09	Bengaluru	Karnataka	South	441	33117050	3976476
STORE010	Store 10	Chennai	Tamil Nadu	South	441	33081273	4242538
STORE005	Store 05	Ludhiana	Punjab	North	429	32736645	4291417
STORE015	Store 15	Bhubaneswar	Odisha	East	431	32643370	4154866
STORE007	Store 07	Pune	Maharashtra	West	441	32072603	3889850

3.5 Management

Q: What are our key KPIs?

```
t13=read.csv("C:/Users/M.QasimSalman/Desktop/MSc/MYSQL/ecommerce/Answers/Management/Q1.csv")
kable(t13)
```

Total.Revenue	Total.Profit	TotalOrders	Total.Customers	Units.Sold	Average.Order.Value	Profit.Margin.Percent
516092560	64511985	6750	999	23720	76458.16	12.5

Q: Where are the biggest opportunities for improvement?